RAJALAKSHMI ENGINEERING COLLEGE

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A2

Started on	Wednesday, 19 August 2026, 8:49 AM
State	Finished
Completed on	Wednesday, 19 August 2026, 8:50 AM
Time taken	1 min 4 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Problem Statement:

Given a sorted array and a value x, the floor of x is the largest element in array smaller than or equal to x. Write divide and conquer algorithm to find floor of x.

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Value for x

Output Format

First Line Contains Integer – Floor value for x

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>
#include <stdlib.h>

// Function to find floor of x using binary search (divide and conquer)
int findFloor(int arr[], int low, int high, int x) {
    // If x is smaller than the smallest element
    if (x < arr[0]) {
        return -1; // No floor exists
    }

    int mid;
    int floorValue = -1; // Initialize as "no floor found"

    while (low <= high) {
        mid = low + (high - low) / 2;

        if (arr[mid] == x) {
            return arr[mid]; // Exact match is the floor
        }
        else if (arr[mid] < x) {
            floorValue = arr[mid]; // Possible floor
            low = mid + 1;         // Search right half
        }
        else {
            high = mid - 1;        // Search left half
        }
    }
    return floorValue;
}

int main() {
    int n;
    if (scanf("%d", &n) != 1 || n <= 0) {
        return 0; // Invalid input
    }

    int *arr = malloc(n * sizeof(int));
    if (!arr) return 0; // Memory allocation failed

    for (int i = 0; i < n; i++) {
        if (scanf("%d", &arr[i]) != 1) {
            free(arr);
            return 0; // Invalid input
        }
    }

    int x;
    if (scanf("%d", &x) != 1) {
        free(arr);
        return 0;
    }

```

	Input	Expected	Got	
✔	6 1 2 8 10 12 19 5	2	2	✔
✔	5 10 22 85 108 129 100	85	85	✔
✔	7 3 5 7 9 11 13 15 10	9	9	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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